



Telit QMI SDK and TQCM

Software User Guide

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1 Applicability Table

Table 1: Applicability Table

Products
1.11.x-y

2 Introduction

2.1 Scope

This document describes how to use Telit QMI SDK and Telit Qualcomm Connection Manager (TQCM) to manage rmnet devices in Linux.

2.2 Audience

This document is intended for Telit customers who want to manage the modem through the rmnet device in Linux using Telit proprietary QMI SDK.

2.3 Contact Information, Support

For technical support and general questions, e-mail:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com

Alternatively, use: <https://www.telit.com/contact-us/>

Product information and technical documents are accessible 24/7 on our website:

<https://www.telit.com>

2.4 Conventions

Note: Provide advice and suggestions that may be useful when integrating the module.

Danger: This information MUST be followed, or catastrophic equipment failure or personal injury may occur.

ESD Risk: Notifies the user to take proper grounding precautions before handling the product.

Warning: Alerts the user on important steps about the module integration.

All dates are in ISO 8601 format, that is YYYY-MM-DD.

2.5 Terms and conditions

Refer to <https://www.telit.com/hardware-terms-conditions/>.

2.6 Disclaimer

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3 Telit QMI SDK and TQCM

3.1 Overview

Telit QMI SDK is a user-space library providing an API to encode/decode QMI messages from/to high level C structs.

Telit Qualcomm Connection Manager (TQCM) is a library providing an API to simplify modem management through QMI requests (e.g. it takes care of the char device related operations): TQCM uses Telit QMI SDK.

TQCM is supporting two different kind of modem devices:

- **QMI devices using the CTL service model**, exposed by the operating system as character devices capable of being managed by standard Linux file system calls (e.g. open, read, poll...). An example of such kind of devices is cdc-wdm as exposed by qmi_wwan kernel driver or wwan0qmi0 as exposed by the wwan subsystem.
- **QRTR device**. An example of such kind of devices is exposed by qrtr and qrtr_mhi kernel modules.

Telit QMI SDK and TQCM are bundled together, being them complementary, and are available in source code.

Telit QMI SDK can be used also with gobinet-like drivers: in this case TQCM should not be used.

Warning: Telit QMI SDK source code is released under NDA. Contact Telit Technical support for further details.

3.2 Package Content

The source code package has the following content:

- **docs**: html documentation.
- **qrtr**: helper library for the qrtr device.
- **sdk**: library for encoding/decoding QMI requests.
- **tqcm**: connection manager.

3.3 Building Telit QMI SDK

Linux

To build Telit QMI SDK dynamic libraries, open a shell in the root of the project and type:

```
$ make
```

Libraries are available inside directories `telit_qmi_sdk/sdk/`, `telit_qmi_sdk/tqcm/`, `telit_qmi_sdk/qrtr/`.

To build Telit QMI SDK and TQCM static libraries, type:

```
$ make static
```

Libraries are available inside directories `telit_qmi_sdk/sdk/`, `telit_qmi_sdk/tqcm/`, `telit_qmi_sdk/qrtr/`.

To build TQCM examples, type:

```
$ make examples
```

To build TQCM static examples, type:

```
$ make examples_static
```

TQCM examples are available inside directory `telit_qmi_sdk/tqcm/examples`.

Android

Copy directory `telit_qmi_sdk` to AOSP directory `vendor/telit`, add packages `libtelitSDK` and `libtqcm` to the build system, then build the AOSP release.

3.4 Telit QMI SDK Usage

Refer to developer documentation available at `html/index.html`.

For code examples with char devices (e.g. `cdc-wdmX`, `wwanXqmiY...`) or QRTR refer to files available at `telit_qmi_sdk/sdk/examples`.

For code examples with gobinet-like devices (e.g. `qcqmiX`) refer to files available at `telit_qmi_sdk/sdk/gobiusbnet_examples`.

3.5 TQCM Usage

Refer to developer documentation available at `html/index.html`.

For code examples refer to files available at `telit_qmi_sdk/tqcm/examples`.

Warning: When used with a QMI character device (e.g. `cdc-wdm` or `wwan0qmi0`), TQCM library can be used only by a single process, since it takes complete control of the device. When, instead, QRTR is used (argument `device_name` set to QRTR when calling `tqcm_init`), TQCM library can be used by different processes.

Data Connection Example

For a data connection example refer to directory

`telit_qmi_sdk/tqcm/examples/data_connection_example/`.

Note: this code supports both QMI devices using the CTL service model and QRTR (specified with the `-d` flag).

The tool is available after building the examples and creates one or more concurrent data connections defined by the data connection configuration file (example available at `telit_qmi_sdk/tqcm/examples/data_connection_example/data_connection.conf`).

For multiple concurrent data connections the flags --qmap, -s and -m are required. The flag -h can be used for details on the tools usage.

Warning: flag --qmap is required for getting the best throughput performance, even for single data connection, so its usage is highly recommended. **Moreover, it is mandatory when the modem is used through PCIe.**

4 Troubleshooting

4.1 Debug Information Retrieval

When contacting customer support for help on troubleshooting issues, please make sure to carefully describe the scenario and provide the following logs:

- Application output with verbosity `LOG_DEBUG` (see function `telit_set_log_level`).

Note: To have better support, follow these guidelines when collecting logs:

- Don't mix different issues in a single log.
- Don't mix different runs of the same issue.
- For long-lasting logs, provide an hint on the timing of the issue.
- Provide a brief description of the scenario captured in the logs.

5 Acronyms and Abbreviations

Table 2: Acronyms and Abbreviations

Acronym	Definition
QC	Qualcomm
MHI	Modem Host Interface
USB	Universal Serial Bus

6 Related Documents

Refer to <https://dz.telit.com/> for current documentation and downloads.

Table 3: Related Documents

S.no	Book Code	Document Title
1	1VV0301371	Telit Modules Linux USB Drivers User Guide
2	1VV0301697	Linux PCIe EP Drivers User Guide

7 Document History

Table 4: Document History

Revision	Date	Changes
10	2024-05-02	Release version update to 1.11.x-y
9	2023-11-16	Release version update to 1.10.x-y
8	2023-09-22	Changed template Release version update to 1.9.x-y
7	2023-03-06	Updated Applicability table, DocumentHistory, Sections Building Telit QMI SDK -> Linux, Data Connection Example, Debug Information Retrieval, and Telit QMI SDK Usage Release version update to 1.8.x-y
6	2023-03-06	Release version update to 1.6.x-y
5	2021-11-19	Added reference to the new data connection example
4	2021-03-26	Added wwan subsystem device names
3	2021-03-26	New document template Minor changes in the paragraphs
2	2021-02-05	Added content related to QRTR and MHI devices
1	2020-05-21	Changed references to build directories Added TQCM usage caveat Added reference to data connection example
0	2020-02-14	First issue

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